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METHOD AND APPARATUS FOR APPLYING A RECTILINEAR BIPHASIC POWER WAVEFORM TO A LOAD

Abstract

A system to deliver therapeutic energy to a patient, the system including a storage capacitor configured to store and release the therapeutic energy, a boost converter circuit coupled to the storage capacitor, and a current flow control circuit coupled to the boost converter circuit and including a plurality of control circuits configured to control a current output from the current flow control circuit in a therapeutic biphasic voltage waveform upon release of the therapeutic energy from the storage capacitor, wherein the therapeutic biphasic voltage waveform includes a ramped increase in voltage from approximately zero volts to a desired therapeutic voltage level over a time interval greater than 1 millisecond and less than a time associated with a phase switch.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/821,203, filed on Aug. 22, 2022, U.S. patent application Ser. No. 16/907,749, filed on Jun. 22, 2020, which is a continuation of U.S. patent application Ser. No. 16/129,864, filed on Sep. 13, 2018, which is a continuation of U.S. patent application Ser. No. 15/647,313, filed on Jul. 12, 2017 and issued as U.S. Pat. No. 10,118,046, which is a continuation of U.S. patent application Ser. No. 15/074,947, filed on Mar. 18, 2016 and issued as U.S. Pat. No. 9,737,722, which is a continuation of U.S. patent application Ser. No. 14/547,865, filed on Nov. 19, 2014 and issued as U.S. Pat. No. 9,289,617, which is a continuation of U.S. patent application Ser. No. 13/967,878, filed on Aug. 15, 2013 and issued as U.S. Pat. No. 8,909,335, which claims the benefit of priority under 35 U.S.C. § 119 (e) to U.S. Provisional Application Ser. No. 61/691,137 filed on Aug. 20, 2012. All subject matter set forth in each of the above referenced applications is hereby incorporated by reference in their entirety into the present application as if fully set forth herein.

BACKGROUND

Field

[0002] Aspects of embodiments relate generally to methods and apparatus for applying a selected energy impulse to a load without exceeding a safe power level. More particularly, aspects of embodiments relate to applying electrical energy impulses to a patient for therapeutic medical purposes. Even more particularly, aspects of embodiments relate to such methods and apparatus as used in heart defibrillators and/or pacing devices.

Discussion of Related Art

[0003] Current defibrillator technology stores electrical energy on a capacitor, a passive energy storage element, preparatory to applying a timed, e.g., 10 msec, rectilinear, biphasic energy impulse of a desired magnitude to a patient. In a known defibrillator, energy is applied as a current from the capacitor through the patient in a first phase, i.e., a first direction, for the first 6 msec of the energy impulse, and then as a current also from the capacitor, but through the patient in an opposite phase to the first phase, i.e., in a second direction opposite to the first direction, for the remaining 4 msec of the energy impulse. In order to accommodate a wide range of patients and operating conditions, especially the voltage droop that occurs as energy is transferred out of the capacitor, the capacitor is charged to a higher level of energy than required to produce the desired energy delivery. The

above-described, known defibrillator incorporates a resistor network into which excess energy is dissipated by diverting a portion of the current from being delivered to the patient when sensors detect that power levels may be dissipated in the patient that exceed safe power levels.

SUMMARY

[0004] According to aspects of an embodiment, a method of applying a rectilinear biphasic electric power waveform to deliver a therapeutic quantity of energy to treat a patient presenting an electrical load is provided. The method comprises storing a quantity of energy substantially equal to and without substantially exceeding the therapeutic quantity of energy on a capacitor, and releasing the stored energy during a first interval in a first direction through the load presented by the patient, in a controlled manner using a boost converter. The method may further comprise releasing the stored energy during a second interval in a second direction through the load presented by the patient. The method may yet further comprise substantially exhausting the stored energy over the first interval and the second interval combined without exceeding a predetermined maximum safe power level when the load presented by the patient is between approximately 25Ω and 200Ω . The method may even yet further comprise releasing a portion of the stored energy from the capacitor into an inductor; releasing the portion of the stored energy from the inductor into the load; and controlling the releasing of the portion of energy into the inductor and into the load in an alternating sequence so as to produce a substantially even flow of energy into the load. According to other aspects of the embodiment, the therapeutic quantity of energy and the first interval are selected to pace a patient whose heart requires pacing impulses. According to yet other aspects of the embodiment, the therapeutic quantity of energy and the first interval are selected to defibrillate a patient whose heart is in fibrillation.

[0005] According to aspects of another embodiment, a system for applying a rectilinear biphasic electric power waveform to deliver a therapeutic quantity of energy to treat a patient presenting an electrical load is provided. The system comprises a capacitor having a rated energy storage capacity substantially equal to the therapeutic quantity of energy, a boost converter constructed and arranged to meter energy out of the capacitor as a substantially constant current while a voltage across the capacitor droops due to decreasing energy stored on the capacitor, and an H-bridge circuit constructed and arranged to apply the current to the patient in the rectilinear biphasic electric power waveform. The system may further comprise a controller that controls for a 10 msec combined first and second interval, and a 6 msec first interval.

[0006] According to aspects of an embodiment, a system to deliver a therapeutic quantity of energy to a patient load is provided. The system comprises capacitor having a rated energy storage capacity substantially equal to the therapeutic quantity of energy, a boost converter coupled with the capacitor and constructed to release energy from the capacitor at a substantially constant current for a time interval, and an H-bridge circuit coupled with the boost converter and constructed to apply the substantially constant current in a biphasic voltage waveform to the patient load.

[0007] According to an embodiment, the boost converter comprises an inductor coupled with the capacitor, a current sensing network, and a solid-state switch coupled between the inductor and the current sensing network. The boost converter may further comprise a controller circuit coupled with the solid state switch and the current sensing network and constructed to cycle the solid state switch. According to other aspects of the embodiment, the current sensing network is constructed to receive a current profile and compare the current profile with a received current from the solid state switch.

[0008] According to an embodiment, the H-bridge circuit comprises a plurality of switches, each of the plurality of switches including a circuit constructed to control the switch and to receive a phase profile having a first phase and a second phase. According to other aspects of the embodiment, the H-bridge circuit may further comprise an inverter coupled with at least two of the plurality of switches to invert the phase profile. At least two switches of the plurality of switches may be configured to be in an open state during the first phase and in a closed state during the second

phase.

[0009] According to an embodiment, the boost converter circuit is further constructed to compensate for voltage droop on the capacitor and variation in the patient load over the time interval. According to an embodiment, the therapeutic quantity of energy and the time interval are selected to defibrillate a patient whose heart is in fibrillation. According to an embodiment, the therapeutic quantity of energy and the time interval are selected to pace a patient whose heart requires pacing impulses.

[0010] According to aspects of an embodiment, a method of delivering a therapeutic quantity of energy to a patient load is provided. The method comprises storing a quantity of energy substantially equal to the therapeutic quantity of energy in a capacitor, releasing the quantity of energy at a relatively constant current during a time interval using a boost converter coupled with the capacitor, and delivering a first portion of the quantity energy in a first direction to the patient load using an H-bridge circuit coupled with the boost converter. According to an embodiment, the method further comprises delivering a second portion of the quantity of energy in a second direction to the patient load using the H-bridge circuit.

[0011] According to an embodiment, releasing the quantity of the stored energy includes transferring energy to an inductor coupled with the capacitor, and sensing the amount of current through a solid state switch coupled between the inductor and a current sensing network. According to an embodiment, releasing the quantity of energy includes cycling the solid state switch using a controller circuit coupled with the solid state switch and the current sensing network. According to an embodiment, releasing the quantity of energy further includes receiving a current profile and comparing the current profile with the amount of current through the solid state switch using the current sensing network.

[0012] According to an embodiment, the H-bridge circuit comprises a plurality of switches and wherein delivering a first portion of the quantity energy in a first direction and a second portion of the quantity of energy in a second direction includes controlling the plurality of switches. According to an embodiment, controlling the plurality of switches includes receiving a phase profile having a first phase and a second phase. According to other aspects of an embodiment, controlling the plurality of switches further includes changing a state of at least 4 switches of the plurality of switches in response to receiving a change in the phase profile from the first phase to the second phase.

[0013] According to an embodiment, releasing the quantity of energy includes compensating for voltage droop on the capacitor and variation in patient load impedance over the time interval. According to an embodiment, the method further comprises determining the therapeutic quantity of energy and the time interval to defibrillate a patient whose heart is in fibrillation. According to an embodiment, the method further comprises determining the therapeutic quantity of energy and the time interval to pace a patient whose heart requires pacing impulses.

[0014] According to aspects of yet another embodiment, a method of maintaining a target power flow from a charge storage device to a patient load while voltage on the charge storage device droops, comprises inserting a boost converter between the charge storage device and the patient load to maintain power flow. The method may further comprise controlling a current delivered by the boost converter so as to compensate for voltage droop on the charge storage device and so as to compensate for variation in patient load impedance over time.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0015] The accompanying drawings are not intended to be drawn to scale. In the drawings, each identical or nearly identical component that is illustrated in various figures is represented by a like

numeral. For purposes of clarity, not every component may be labeled in every drawing. In the drawings:

- [0016] FIG. 1 is a schematic drawing of a circuit for delivering a rectilinear biphasic electric power waveform to deliver a therapeutic quantity of energy to treat a patient presenting an electrical load;
- [0017] FIG. 2 is a graph of electrical waveforms produced by the circuit of FIG. 1 over a period of time with a patient load of 25 Ω ;
- [0018] FIG. 3 is a graph of electrical waveforms produced by the circuit of FIG. 1 over a period of time with a patient load of 50 Ω ;
- [0019] FIG. 4 is a graph of electrical waveforms produced by the circuit of FIG. 1 over a period of time with a patient load of 100 Ω ;
- [0020] FIG. 5 is a graph of electrical waveforms produced by the circuit of FIG. 1 over a period of time with a patient load of 150 Ω ;
- [0021] FIG. 6 is a graph of electrical waveforms produced by the circuit of FIG. 1 over a period of time with a patient load of 200 Ω ;
- [0022] FIGS. 7A-F illustrate various current profiles and phase profiles that may be used with the circuit of FIG. 1 to deliver a variety of different defibrillating waveforms to the body of a patient;
- [0023] FIG. 8 is a schematic drawing of a drive circuit for switches used in an H-bridge sub-circuit of the circuit of FIG. 1;
- [0024] FIG. 9 is a graph of electrical waveforms produced by the circuit of FIG. 1 with a patient load of 25 Ω , when programmed for pacer mode; and
- [0025] FIG. 10 is a graph of electrical waveforms produced by the circuit of FIG. 1 with a patient load of 300 Ω , when programmed for pacer mode.

DETAILED DESCRIPTION

[0026] This invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

[0027] As noted in the BACKGROUND section, defibrillators are devices that deliver a desired quantity of energy to a patient without exceeding a safe power level. Energy is simply power delivered to a load over a period of time:

[00001] $Pt = E$; or $P = E / t$; [0028] where P represents power in Watts, E represents energy in Joules, and t represents the period of time in seconds over which the energy is delivered. When electrical energy is dissipated in a simple resistive load, that is, one which resists a flow of electrical current when a voltage is applied, power may be expressed in terms of the voltage applied to the load, voltage being a measure of electrical pressure across the load, and current through the load, current being a measure of movement of charge through the load. Electrical power is:

[00002] $P = VI$; or $P = I^2 R$; or $P = V^2 / R$; [0029] where V represents voltage in Volts, I represents current in Amperes, and R represents the resistance of the load in Ohms.

[0030] Defibrillators store the desired quantity of electrical energy on a capacitor, as a charge. Storing a charge on a capacitor causes a voltage to appear across the terminals of the capacitor. When a user of a defibrillator applies a therapeutic shock to a patient, the electrical energy stored on the capacitor is released through the patient, whose body provides substantially a simple resistive load in which the energy is dissipated. As the capacitor supplies energy to the load, the charge on the capacitor decreases, and so the voltage appearing across the capacitor also decreases. As voltage decreases, or sags, the current driven through the load also decreases. Applying any of the definitions of electrical power given above, it is observed that the power, $P_{sub.0}$, delivered by

the capacitor at the beginning of a therapeutic shock of a defined magnitude, E.sub.0, is greater than the power, P.sub.N, delivered by the capacitor at the end of the therapeutic shock because the voltage on the capacitor sags as the charge on the capacitor is depleted by supplying current to the patient.

[0031] Conventionally, in order to accommodate the voltage sag, while delivering a constant, desired maximum power level until the desired energy impulse has been delivered, the size of the capacitor is selected to provide the desired energy impulse to a worst-case load at the end of the energy impulse. For these purposes, a worst-case load may be considered to be one at a lower end of an expected resistance range, since such a load will require a larger current to maintain a constant power level during the energy impulse. Such a design requires a capacitor that, when charged to a level that yields the desired energy impulse, dissipates in the patient a power level in excess of that desired during the initial portion of the energy impulse. As previously explained, during times of excess power delivery, the excess energy is simply dissipated into resistors so as to reduce to desired maximum levels the power delivered to the load, i.e., the patient, which both wastes power and necessitates the use of a capacitor whose rated energy storage capacity is greater than the maximum energy delivery requirement, since energy is dumped into the dissipation resistors and not recovered or otherwise put to therapeutic use.

[0032] Using a capacitor whose rated energy storage capacity is greater than the maximum energy delivery requirement is disadvantageous from several perspectives. For a given capacitor technology, greater storage capacity requires greater size and/or weight. A physically larger capacitor is undesirable, particularly for use in portable equipment, because equipment must be built larger and is more difficult to transport. Size and weight factors can prove prohibitive for equipment meant to be worn by, transported with, or carried by, a patient who themselves may not be fully ambulatory. Moreover, energy that is wasted, yet must be stored on the capacitor as described above, adds to the charging time and the performance characteristics required of the charging circuit which places the energy on the capacitor.

[0033] For example, in the conventional defibrillator described in the BACKGROUND section in which excess energy is dissipated into dissipation resistors, a capacitor having a minimum required energy rating of approximately 381 Joules is used. Under favorable conditions for maximum energy shock (i.e., a 200 Joule setting into a patient presenting an impedance of 161 Ω), approximately 69% of the capacitor's minimum required energy rating is delivered to the patient. For higher impedance patients, energy utilization drops off slightly to 67% for a patient presenting an impedance of 175 Ω , and to 63% for a patient presenting an impedance of 200 Ω . The drop off in energy utilization is more severe for lower impedance patients (e.g., 37% for a patient presenting a 25 Ω impedance, and 21% for a patient presenting a 15 Ω impedance), primarily due to energy dissipated in the dissipation resistors. One of the physically smallest capacitors validated for use in such a conventional defibrillator weighs approximately 10 oz (283.5 grams) and has a volume of approximately 20 in.sup.3 (327.7 cm.sup.3).

[0034] According to aspects of embodiments, a boost converter is employed to control and regulate the delivery of a constant current, resulting in a constant power dissipation level during the delivery of a desired energy impulse. In brief summary, a boost converter transfers energy in very short bursts compared to the time for delivering the total desired energy impulse, first from the capacitor to an inductor, which stores the energy as a substantially constant current, and then from the inductor to the patient. Because the current delivered to the patient by the inductor is substantially constant due to the intrinsic electrical characteristics of inductors which tend to resist a change to current through them, a constant, maximum desired power level is dissipated in the patient, in accordance with the definitions of electrical power given above. A boost converter circuit of a defibrillator incorporating aspects of embodiments is now described in greater detail.

[0035] First, the basic boost converter circuit is described in connection with FIG. 1. The boost converter circuit, **100**, provides a substantially constant current at its output node, **101**, when that

node is connected to a load, **102**. The circuit, **100**, includes a storage capacitor, **103**, in which the energy for the desired impulse is held until a discharge into the patient load is triggered; an inductor, **104**, connected to receive a current from the storage capacitor, **103**, when the discharge is triggered; a diode, **105**, to protect against a reversal of the current discharge; and, optionally a smoothing capacitor, **106**; as well as control elements enumerated below. A charge circuit, such as a battery or other DC power source (not shown) is coupled to the storage capacitor **103**, for example via relays, to provide energy to the storage capacitor **103**. A terminal of the storage capacitor **103** is electrically coupled to a first terminal of the inductor **104**, which in FIG. 1 is modeled as an inductor **1041** coupled in series with a resistor **104r**. The second terminal of the inductor **104** is electrically coupled to the anode of the diode **105**, with the cathode of the diode **105** being electrically coupled to a first terminal of the optional smoothing capacitor **106** and to the output node **101**.

[0036] According to the capacitor energy equation, $E = \frac{1}{2}CV^2$, an exemplary capacitor, **103**, of 270 μF , as shown in FIG. 1, charged to about 1218 V would store 200 Joules for the exemplary therapeutic shock. To deliver 200 Joules over a 10 msec impulse requires delivering a substantially constant, instantaneous power of 20 kW to the load, **102**. If the load, **102**, is 25 Ω , then the power equation, $P = I^2R$, calls for a current of 28 A, while a load, **102**, of 200 Ω calls for a current of 10 A. Inductor, **104**, is of a size to prevent substantial current droop while delivering a desired power level to the patient load, **102**. A 1 mH inductor, **104**, as shown in FIG. 1, produces the desired result, as illustrated below in FIGS. 2-6. In accordance with one embodiment, the inductor **104** may be an unsaturable 1 mH Litz wire air-core coil dimensioned to optimize self-inductance.

[0037] While 100% utilization of the capacitor energy storage capability is the theoretical goal, practical circuit elements, which have real losses associated with them, achieve somewhat lower utilization rates, per the Table I, below. The simulations presented in FIGS. 2-6, and discussed below, assume capacitor, **103**, has a capacitance of 270 μF , and an initial stored energy of 305 Joules.

TABLE-US-00001 TABLE I Therapeutic 10 msec Initial Patient Shock Continuous Energy
Impedance Energy Power Usage 25 Ω 249 Joules 24.9 kW 81% 50 Ω 255 Joules 25.5 kW 83%
100 Ω 236 Joules 23.6 kW 77% 150 Ω 221 Joules 22.1 kW 72% 200 Ω 210 Joules 21.0 kW 69%

[0038] By comparison to a conventional defibrillator using a storage capacitor having a minimum energy rating of 381 Joules, embodiments of the present invention permit the use of a storage capacitor having an approximately 20% lower minimum energy rating (e.g., 305 Joules) while providing a similar amount of energy to the patient. As a result, the size and weight of the storage capacitor **103** used with embodiments of the present invention may be reduced by approximately 20% relative to storage capacitors used in a conventional defibrillator. Further efficiencies of size and cost are provided by eliminating the need for dissipation resistors and their associated shunting devices used in conventional defibrillators, as well as any of the thermal management features needed to dissipate the heat generated therefrom.

[0039] A specialized controller circuit, **107**, modeled for convenience as a UC3842 current mode PWM controller, has a control output connected to a control input of a high-voltage and high-current, solid-state switch, **108** that is coupled between the second terminal of the inductor **104** and a current sensing network **109**. The solid state switch **108** may be an IGBT as shown in FIG. 1, or another type of a high-voltage and high current solid state switch, such as a thyristor. It should be appreciated that embodiments of the present invention are not limited to the use of a particular type of PWM controller or to a particular type of high-current solid state switch, as other types of controller circuits, and other types of high-current switches may alternatively be used. Current drawn through the switch **108** is measured by the current sensing network, **109**; compared to a desired current profile, **110**; and, the result is provided as an input to the controller circuit, **107**. Since, as explained above, current is directly related by a square law to instantaneous power, controlling for a desired current also controls for the desired instantaneous power level.

[0040] The load presented by the patient, **102**, is connected to the output node, **101**, through an H-bridge structure which causes current to flow through the patient in a desired direction at a desired time. The H-bridge includes four H-bridge switches **111**, with each H-bridge switch **111a**, **111b**, **111c**, **111d** including a respective switching transistor **116a**, **116b**, **116c**, **116d** and a respective control circuit **117a**, **117b**, **117c**, **117d** associated with each. The switching transistors can be insulated-gate bipolar transistors (IGBTs), metal-oxide semiconductor field-effect transistors (MOSFETs), silicon-controlled rectifiers (SCRs) or such other high-current switching devices as may be available. In the exemplary, illustrative embodiment, for modeling purposes only, an oscilloscope, **112**, having a channel A input, **112A**, and a channel B input, **112B**, has been included. Channel A, **112A**, monitors the current impulses passed through the switch, **108**, and channel B, **112B**, monitors the voltage across the patient load, **102**. The traces produced by channels A and B, **112A** and **112B**, are shown in FIG. 2, which is next referred to in an explanation of the operation of the circuit of FIG. 1.

[0041] The circuit of FIG. 1 operates as follows to provide a 200-Joule defibrillation shock to a patient load, **102**, of 25Ω , as illustrated in FIG. 2. The storage capacitor, **103**, is first charged up with about 200 Joules of electrical energy, by a charging circuit (not shown). There are small, parasitic losses due to parasitic resistances throughout the circuits which deliver the charge to the patient, including parasitic resistances in the inductor, **104r**, and elsewhere. If the parasitic resistances are negligible, then no more than about 200 Joules need be stored on the capacitor; however, if the parasitic resistances are non-negligible, then the storage capacitor, **103**, should hold a small excess above the desired 200 Joules of electrical energy, the excess being sufficient to just account for the energy dissipated in the parasitic resistances under worst-case conditions. It should be appreciated that other shock energies (i.e., other than 200 Joules) may be provided, as known to those skilled in the art.

[0042] According to one embodiment, operation begins with the solid-state switch, **108**, open, and each of the H-bridge switches, **111a-d**, open. When a therapeutic shock is triggered, a pair of the H-bridge switches, e.g., **111a** and **111c**, is closed, initiating current through the patient load **102**. Current then builds up in the inductor, **104**. As shown in FIG. 2, indicated by line, **200**, current through switch, **108**, is zero during this initial period, **201**. Next, during period, **202**, the controller circuit, **107**, begins cycling switch, **108**, on and off, thereby allowing current through switch, **108**. When the switch, **108**, is closed and the controller circuit, **107**, detects that the desired current or higher is flowing through the switch, **108**, it provides a control signal to the solid-state switch, **108**, to again open the switch, **108**, allowing current through the inductor, **104**, to the patient load, **102**. At regular intervals, the controller, **107**, closes the switch, **108**, and checks for the current to build up to the desired level, at which point the control signal again opens the switch, **108**.

[0043] During each cycle, during period, **202**, when the controller circuit, **107**, determines from the output of the current sensing network, **109**, that the correct current level has been reached or exceeded, a control signal is applied to the solid-state switch, **108**, to open the switch, allowing current through the inductor, **104**, and the patient load, **102**, from the energy stored on storage capacitor, **103**. As current is initiated through the patient, a voltage, indicated in FIG. 2 by line **210**, appears across the patient that causes the current in the inductor, **104**, to begin to decay, and so the controller circuit, **107**, again closes solid-state switch, **108**, to begin the cycle again by building up the current stored in the inductor. By repeating the forgoing cycle many times during the therapeutic shock, energy stored on the storage capacitor, **103**, is metered out to the patient without ever exceeding the maximum allowable power dissipation level in the patient. According to some embodiments, it has been found that the desired waveform to be applied to the patient reverses polarity after an interval. Accordingly, the H-bridge switches are controlled by a desired phase profile, **120**, to open the closed pair of switches, **111a** and **111c**, and close the open pair of switches, **111b** and **111d**, at about 6 msec into the therapeutic shock cycle, reversing the polarity of the applied shock, **211**. It should be noted that switches **111a** and **111c** are opened prior to closing

switches **111b** and **111d** to avoid short-circuiting the H-bridge structure. The magnitude of the current applied to the patient load, **102**, (and the resulting voltage across the patient load, **102**) does not substantially change during the polarity reversal, **211** and **213**.

[0044] As shown in FIGS. **3**, **4**, **5**, and **6**, the operation is similar for patients presenting resistance values of 50 Ω , 100 Ω , 150 Ω , and 200 Ω . The variation in the load, **102**, results in different damping characteristics for the therapeutic shock waveforms, i.e., the overall shape of the waveform, and also results in different patient voltages, such that the 200 Joule impulse is applied as desired. In each of FIGS. **3**, **4**, **5**, and **6**, reference numerals indicating corresponding elements to elements of FIG. **2** correspond, except for the hundreds place, which corresponds to the FIG. number. For example, FIG. **2**, line **200**, corresponds to FIG. **3**, line **300**, but for a different patient load, **102**.

[0045] As shown in FIG. **2**, for a patient load, **102**, of 25 Ω , the absolute value of patient voltage **210** varies between a peak of about 1.5 kV and 500 V. For a patient load, **102**, of 50 Ω , the absolute value of patient voltage **310** has a much flatter shape, as shown in FIG. **3**. It hits a peak of about 2 kV but remains for most of the impulse at about 1.2 kV, finally tapering down to just under 1.0 kV. As shown in FIG. **4**, for a patient load, **102**, of 100 Ω , the absolute value of patient voltage **410** has an even flatter shape. It also hits a peak of about 2 kV but remains for most of the impulse at about 1.5 kV, finally tapering down to about 1.2 kV. For a patient load, **102**, of 150 Ω , the absolute value of patient voltage **510** has a quite flat shape, as shown in FIG. **5**. It hits a peak of about 2 kV but remains for nearly the entire impulse at about 1.8 kV. For a patient load, **102**, of 200 Ω , the absolute value of patient voltage **610** has a quite flat shape. It hits a peak of about 2 kV but remains for nearly the entire impulse at about 1.8 kV, as shown in FIG. **6**.

[0046] In practical systems, the preference is to deliver substantially constant energy to the patient during a period of time. Thus, if a 200 Joule therapeutic shock is desired to be delivered in a 10 msec period, the controller circuit, **107**, is designed or programmed to obtain a current level in the inductor, **104**, that delivers 20 J/msec. The controller circuit, **107**, in connection with the current sensing network, **109**, the desired current profile, **110**, and the solid state switch, **108**, forms a feedback loop that controls and maintains the 20 J/msec level, or such other level or waveform as desired. Except for parasitic losses, explained below, the storage capacitor, **103**, need only have a rated energy storage capacity of 200 Joules, since no excess energy is dumped and it is desired to leave no residual energy in the storage capacitor, **103**, after the therapeutic shock has completed.

[0047] FIGS. **7A-F** illustrate the manner in which the boost converter circuit **100** of FIG. **1** may be used to control the shape and/or phase of the defibrillating waveform applied to the patient. As described previously with respect to FIG. **1**, current drawn through the switch **108** is measured by the current sensing network **109**, compared to a desired current profile **110**, and provided as an input to the controller circuit **107**. In the circuit illustrated in FIG. **1**, this comparison is performed by an operational amplifier **118** configured as a comparator. By controlling the shape and amplitude of the desired current profile **110**, a waveform of a desired amplitude and desired shape may be delivered to the patient load **102**.

[0048] For example, as shown in FIG. **7A**, a current profile **110** having a step impulse that corresponds to a current level of approximately 22 A and that gradually increases to a level corresponding to approximately 40 A may be used to provide the current and voltage waveforms depicted in FIGS. **3-6**. The current profile **110** is approximately 10 msec in duration and includes an initial period or region **110a** where the current is zero (corresponding to periods **301**, **401**, **501**, and **601** in FIGS. **3-6**), followed by a step impulse (region **110b**) corresponding to approximately 22 A. At approximately 1 msec, the current profile **110** linearly increases (region **110c**) to a value corresponding to approximately 24 A. The current profile **110** remains at a level corresponding to about 24 A for about 2 msec (region **110d**), where it then linearly increases (region **110e**) to a level corresponding to approximately 40 A, after which the current profile returns to a zero current level (region **110f**) at approximately 10 msec. The overall shape of the current profile **110** used to

generate the current and voltage waveforms depicted in FIGS. 3-6 is shown in each respective figure by the respective envelope or profile **303**, **403**, **503**, and **603** of the current through the switch **108**. Differences in the voltage waveforms **310**, **410**, **510**, and **610** applied to the patient in FIGS. 3-6 are primarily due to differences in the patient load **102**. The overall shape of the current profile **110** used to generate the current and voltage waveforms depicted in FIG. 2 is shown in FIG. 2 by the envelope **203** of the current **200** through the switch **108**.

[0049] The phase of the defibrillating waveform that is applied to the patient may be controlled by the desired phase profile **120** provided to each of the H-bridge switches **111**. For example, FIG. 7B illustrates a desired phase profile **120** that may be used to control the phase of the waveforms depicted in FIGS. 2-6. As shown, the phase profile **120** initially assumes a high state at time zero (or before) followed by a change to a low state at 6 msec. In the circuit depicted in FIG. 1, each of H-bridge switches **111a** and **111c** receives the phase profile **120**, while each of H-bridge switches **111b** and **111d** receives an inverted version of the phase profile **120**. The high state of the phase profile **120** operates to fully close each of H-bridge switches **111a** and **111c** during the initial 6 msec, and to maintain each of switches **111b** and **111d** in a fully open position. At approximately 6 msec, the level of the phase profile **120** changes, thereby fully opening H-bridge switches **111a** and **111c**, and fully closing switches **111b** and **111d**, thereby reversing the polarity of the delivered voltage waveform as shown in each of FIGS. 2-6. The presence of the inverter **115** serves to not only invert the phase profile **120**, but to delay the signal provided to each of H-bridge switches **111b** and **111d** to help ensure that these switches are not closed until after switches **111a** and **111c** have opened. If necessary, additional delays could be provided. Where the switching transistors **116a**, **116b**, **116c**, and **116d** used in each of the H-bridge switches **111a**, **111b**, **111c**, **111d** are capable of operating in a linear mode, the high state and the low state of the phase profile **120** should be such that the switching transistors are either fully conducting (on) or fully non-conducting (off) to avoid thermal destruction.

[0050] FIG. 7C illustrates an alternative desired current profile **110** that may be used with the boost converter circuit **100** of FIG. 1 to generate a biphasic voltage waveform that increases asymptotically from a zero value to a desired voltage level (e.g., to approximately 2 kV in amplitude for a 200 Ω patient) over the initial portion of each phase. When combined with a phase profile **110** similar to that illustrated in FIG. 7B, the boost converter circuit **100** may provide a defibrillating voltage **710** to the body of the patient similar to that shown in FIG. 7D. As shown in FIG. 7D, the biphasic voltage waveform **710** is approximately 10 msec in duration and switches phase at approximately 6 msec. During the first few milliseconds of each phase, the voltage **710** applied to the body of the patient rises asymptotically to an amplitude of about 2000 V (for a 200 Ω patient). To achieve the shape of the voltage waveform **710** shown in FIG. 7D, the smoothing capacitor **106** (FIG. 1) may be omitted or set to a value of zero. Such a ramped asymptotic voltage waveform as shown in FIG. 7D may reduce the amount of trauma to the patient's heart during defibrillation, by avoiding the step impulse in voltage shown in each of FIG. 2-6.

[0051] FIGS. 7E and 7F illustrate alternative phase profiles that may be used with the current profile **110** discussed above with respect to FIG. 7A. For example, FIG. 7E illustrates a desired phase profile **120** having two opposing phases over a 10 msec duration, in which the opposing phases are substantially similar in duration (i.e., about 5 msec each). Such a phase profile may be used to balance the amount of charge delivered to the patient's heart in each direction and thereby potentially reduce the trauma to the patient's heart. FIG. 7F illustrates yet an alternative phase profile that may be used to provide a monophasic defibrillating shock to the body of the patient.

[0052] It should be appreciated that a variety of different current profiles **110** and phase profiles **120** may be used with the boost converter circuit **100** of FIG. 1 to generate a corresponding variety of different defibrillating waveforms, each having a different shape and/or amplitude and/or phase. For example, the current profile **110** could include a simple exponential waveform or a damped sine wave, and the phase profile **120** could be monophasic, biphasic, triphasic, or otherwise.

Accordingly, where it is determined that a particular shape, amplitude, phase, sequence of phases, or all of the above is particularly effective, reference waveforms may be generated and used as desired current profile **110** and desired phase profile **120** to achieve the desired resultant defibrillating waveform. Although not depicted in FIG. **1**, each of the desired current profile **110** and the desired phase profile **120** may be stored in a memory, and provided, for example, by a processor of the defibrillator to comparator **118** and each of H-bridge switches **111** to vary the shape, amplitude, or phase of the defibrillating waveform applied to the patient as desired.

[0053] Referring back to FIG. **1**, again, within each H-bridge switch, **111a-d**, is a driver circuit (i.e., **117a**, **117b**, **117c**, and **117d**). These circuits are feedback systems illustrated in greater detail in FIG. **8**. Each driver circuit is configured to control a switching transistor (i.e., **116a**, **116b**, **116c**, and **116d**, respectively) to pass a current up to a controlled maximum level above which the current is clipped.

[0054] In each driver circuit is an operational amplifier, **802**, connected to receive a control signal, DRIVE, that is based upon the desired phase profile **120** and produce an output, GATE, which turns a switching transistor (FIGS. **1**, **116a**, **116b**, **116c**, and **116d**) on and off as required. A second operational amplifier, **804**, is connected to sense current through each switching transistor (FIGS. **1**, **116a**, **116b**, **116c**, and **116d**), and control an input to operational amplifier, **802**, so a voltage at output, GATE, turns the switching transistor on up to a desired maximum current through the switching transistor. In the embodiment depicted in FIG. **1**, each of the switching transistors **116a**, **116b**, **116c**, and **116d** is an IGBT that may be operated in a non-linear mode as a two state (i.e., on or off) switch, or in a linear mode as a voltage controlled current source. Where the circuit **100** is used to provide therapeutically effective amounts of energy sufficient for defibrillation, the switching transistors **116a**, **116b**, **116c**, and **116d** would typically be operated in the non-linear mode (e.g., as a two state switch) to avoid thermal destruction. However, by adjusting the timing defined by controller circuit, **107**, the desired current profile, **110**, and the desired phase profile, **120**, the circuit of FIG. **1** can be programmed to perform pacing, as well as defibrillation. For example, by appropriately controlling the timing defined by the controller circuit **107** and the desired current profile **110**, and at energies typically used for pacing, the smoothing capacitor **106** can substantially correspond to a DC power supply of a desired voltage. By then providing a suitable phase profile to pairs of H-bridge switches (i.e., **111a** and **111c**, **111b** and **111d**), the switching transistors of a respective pair of H-bridge switches may be operated in their linear mode to provide a desired pacing waveform.

[0055] FIGS. **9** and **10** illustrate a phase profile **120** that may be provided to pairs of H-bridge switches, **111a** and **111c** or **111b** and **111d**, to deliver suitable pacing waveforms **902**, **1002**, to a patient load **102**, when the timing defined by the controller circuit **107** and the current profile **110** have been appropriately programmed. In each of FIGS. **9** and **10**, the timing of the controller circuit **107** and the current profile **110** have been programmed so that smoothing capacitor **106** effectively operates as a 50 V D.C. voltage source. As illustrated in FIG. **9**, a voltage level of approximately 27 mV provided to the DRIVE input of driver circuits **117A** and **117C** is sufficient to deliver an 8 mA, 200 mV pacing pulse into a patient **102** presenting a 25 Ω load. As illustrated in FIG. **10**, a voltage level of approximately 465 mV is sufficient to deliver an 140 mA, 42 V pacing pulse into a patient **102** presenting a 300 Ω load. By appropriately controlling the phase profile **120**, pacing pulses ranging from a few milliamps to two hundred milliamps or more may be provided to a patient, using the same circuit topology as that used to deliver a defibrillating shock.

[0056] It should be appreciated that although embodiments of the present invention have been primarily described with respect to defibrillation and pacing, they may also be used to deliver other types of therapeutic waveforms to the body of a patient in which the energy delivered is between the ranges of energy typically used for pacing or defibrillation. For example, pacing pulses typically range from a few mA to approximately 200 mA, and defibrillation pulses typically range from about 1 A to about 35-40 A. Between these ranges of current exists a wide spectrum of

energies that may be applied to the body of a patient for a variety of therapeutic purposes, for example, to perform charge bumping of a patient's heart, etc. Accordingly, by varying the timing of the controller circuit, the current profile **110**, and the phase profile **120**, embodiments of the present invention may be used to tailor one or more of the shape, the voltage, and the current of a therapeutic waveform to be applied to the body of a patient.

[0057] Having thus described several aspects of at least one embodiment of this invention, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this disclosure, and are intended to be within the scope of the invention. Accordingly, the foregoing description and drawings are by way of example only.

Claims

1. (canceled)

2. An apparatus for providing electrotherapy to a patient, the system comprising: a capacitor configured to store and release energy for the electrotherapy; one or more switches configured for adjusting an electrotherapy waveform being delivered to the patient; a boost converter circuit coupled with the capacitor, the boost converter circuit comprising at least one inductor configured to store energy provided by the capacitor; a current sensing circuit coupled to the at least one inductor and configured to sense electrical signals indicative of a current being delivered to the patient; and a controller circuit coupled with the current sensing circuit, the controller circuit configured to control the one or more switches in adjusting the electrotherapy waveform being delivered to the patient.

3. The apparatus of claim 2, wherein the electrotherapy waveform comprises at least one of: a current waveform, a voltage waveform and a power waveform.

4. The apparatus of claim 2, wherein the at least one inductor comprises an unsaturable air-core inductor.

5. The apparatus of claim 4, wherein the at least one inductor comprises a 1 mH inductor.

6. The apparatus of claim 4, wherein the at least one inductor comprises a Litz wire air-core coil.

7. The apparatus of claim 4, wherein the at least one inductor comprises dimensions for optimization of self inductance.

8. The apparatus of claim 2, wherein the one or more switches are coupled between the at least one inductor and the current sensing circuit.

9. The apparatus of claim 8, wherein current flowing through the one or more switches is measured by the current sensing circuit.

10. The apparatus of claim 2, comprising at least one polarity control circuit.

11. The apparatus of claim 2, wherein the at least one polarity control circuit comprises a plurality of switches.

12. The apparatus of claim 2, wherein the at least one polarity control circuit comprises an H-bridge.

13. The apparatus of claim 2, wherein the at least one polarity control circuit is configured to apply the electrotherapy waveform to the patient.

14. The apparatus of claim 2, wherein the boost converter circuit is configured to increase a voltage of the electrotherapy waveform being delivered to the patient in response to a drop in the voltage.

15. The apparatus of claim 2, wherein the boost converter circuit is configured to increase a voltage of the electrotherapy waveform being delivered to the patient in adjusting the therapeutic waveform being delivered to the patient.

16. The apparatus of claim 2, wherein the controller circuit comprises a PWM controller.

17. The apparatus of claim 4, wherein the boost converter circuit is configured to release energy from the capacitor at a substantially constant current.

18. The apparatus of claim 4, wherein the electrotherapy waveform is biphasic.

19. The apparatus of claim 4, wherein the electrotherapy waveform is a rectilinear biphasic waveform.

20. The apparatus of claim 4, wherein the electrotherapy waveform comprises a defibrillation waveform.

21. The apparatus of claim 4, wherein the electrotherapy waveform comprises a pacing.
